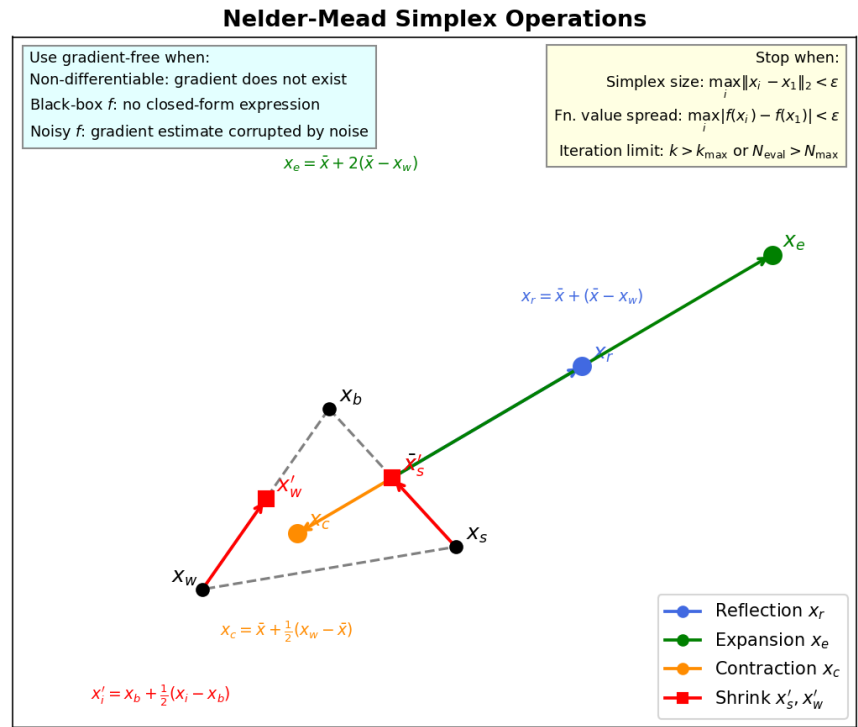


Newton on KKT	KKT Analytical
$\min f(x) = x_1^2 + 5x_2^2, \text{ s.t. } g(x) = 4 - x_1 - x_2 \leq 0$	$\min f(x) = x_1^2 + 5x_2^2, \text{ s.t. } g(x) = x_1 + x_2 - 4 \leq 0$
	Slack: $g(x) - s^2 = 0$, so $l(x, \lambda, s) = f(x) - \lambda(g(x) - s^2)$
	$\frac{\partial l}{\partial s} = 2\lambda s = 0 \Rightarrow \lambda = 0$ (inactive) or $s = 0$ (active)
	Test: $X_{\text{unc}} = (0, 0)$; $g(X_{\text{unc}}) = -4 < 0 \therefore$ infeasible $\Rightarrow \lambda \neq 0 \xrightarrow{2\lambda s=0} s = 0$, constraint active
$l(x, \lambda) = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$ (same l , implicit in $\nabla_x l$)	$l(x, \lambda) = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$ (set $s = 0$)
Assert $r(x, \lambda) = 0$ and J ; note $J = \begin{bmatrix} H_l & -\nabla_x g \\ \nabla_x g^\top & 0 \end{bmatrix}$:	Same r and J arrived at from $\nabla_x l = 0$ and $g = 0$:
$r = \begin{bmatrix} \nabla_x l \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$	$r = \begin{bmatrix} \nabla_x l \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$
Init: $z_0 = (0, 0, 0)^\top$; evaluate r at z_0 :	(no iterate — constraint known active from test above)
$r(z_0) = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix}_{z_0=(0,0,0)} = \begin{bmatrix} 2(0) - 0 \\ 10(0) - 0 \\ 0 + 0 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -4 \end{bmatrix}$	
Newton step: $J \Delta z = -r(z_0)$, i.e. $-r(z_0) = (0, 0, 4)^\top$:	$\nabla_x l = H_f x - \lambda \nabla_x g = 0 \Rightarrow x^*(\lambda) = H_f^{-1} \lambda \nabla_x g$:
$J \Delta z = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \end{bmatrix} = 0 \Rightarrow x^*(\lambda) = \begin{bmatrix} \lambda/2 \\ \lambda/10 \end{bmatrix}$
Row 1: $2\Delta x_1 - \Delta \lambda = 0 \Rightarrow \Delta x_1 = \frac{\Delta \lambda}{2}$ Row 2: $10\Delta x_2 - \Delta \lambda = 0 \Rightarrow \Delta x_2 = \frac{\Delta \lambda}{10}$	
Row 3: $\frac{\Delta \lambda}{2} + \frac{\Delta \lambda}{10} = 4 \Rightarrow 6\Delta \lambda = 40 \Rightarrow \Delta \lambda = \frac{20}{3}$	Sub $x^*(\lambda)$ into $g = 0$: $\frac{\lambda}{2} + \frac{\lambda}{10} = 4 \Rightarrow 10\lambda + 2\lambda = 80 \Rightarrow \lambda = \frac{20}{3}$
$\Delta x_1 = \frac{10}{3}, \quad \Delta x_2 = \frac{2}{3}; \quad z_1 = z_0 + \Delta z = \left(\frac{10}{3}, \frac{2}{3}, \frac{20}{3}\right)$	$x^* = \left(\frac{10}{3}, \frac{2}{3}\right)^\top$
$f(x^*) = \left(\frac{10}{3}\right)^2 + 5\left(\frac{2}{3}\right)^2 = \frac{100}{9} + \frac{20}{9} = \frac{40}{3}$	$f(x^*) = \left(\frac{10}{3}\right)^2 + 5\left(\frac{2}{3}\right)^2 = \frac{100}{9} + \frac{20}{9} = \frac{40}{3}$



BFGS quasi-Newton $p_0 = -\tilde{\nabla}^2 f(X_0)^{-1} \nabla f(X_0)$, $B_0 = I$
 $f(x_1, x_2) = x_1^2 + 5x_2^2$, $X_0 = (3, 1)^\top$

$$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \Rightarrow \nabla f(X_0) = \begin{bmatrix} 2 \cdot 3 \\ 10 \cdot 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$\text{Candidate: } \hat{X}_1 = X_0 + \alpha p_0 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \alpha \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 3 - 6\alpha \\ 1 - 10\alpha \end{bmatrix}$$

$$\phi(\alpha) = (3 - 6\alpha)^2 + 5(1 - 10\alpha)^2$$

$$= 9 - 36\alpha + 36\alpha^2 + 5(1 - 20\alpha + 100\alpha^2)$$

$$= 14 - 136\alpha + 536\alpha^2$$

$$\frac{d\phi}{d\alpha} = -136 + 1072\alpha = 0 \Rightarrow \alpha_0 = \frac{136}{1072} \approx 0.1269$$

$$X_1 = X_0 + \alpha_0 \nabla f(X_0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - 0.1269 \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 2.239 \\ -0.269 \end{bmatrix}$$

BFGS vectors:

$$s_0 = -\alpha_0 \nabla f(X_0) = -0.1269 \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -0.7614 \\ -1.269 \end{bmatrix}$$

$$y_0 = \nabla f(X_1) - \nabla f(X_0) = \begin{bmatrix} 2 \cdot 2.239 \\ 10 \cdot (-0.269) \end{bmatrix} - \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -1.522 \\ -12.689 \end{bmatrix}$$

Newton — unconstrained $p_0 = -H_f(X_0)^{-1} \nabla f(X_0)$
 $f(x_1, x_2) = x_1^2 + 5x_2^2$, $X_0 = (3, 1)^\top$

$$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \Rightarrow \nabla f(X_0) = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$H_f = \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix} \Rightarrow H_f^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix}$$

$$p_0 = - \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix} \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$$

$$\text{Line search: } \hat{X}_1 = \begin{bmatrix} 3 - 3\alpha \\ 1 - \alpha \end{bmatrix}$$

$$\phi(\alpha) = (3 - 3\alpha)^2 + 5(1 - \alpha)^2 = 14 - 28\alpha + 14\alpha^2$$

$$\frac{d\phi}{d\alpha} = -28 + 28\alpha = 0 \Rightarrow \alpha_0 = 1$$

$$\text{Update: } X_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + 1 \cdot \begin{bmatrix} -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$f(X_1) = 0$ — one step, exact (quadratic f , $\alpha = 1$ always)

(h) [3 marks] The particle swarm update is:

$$x_{k+1}^{(i)} = x_k^{(i)} + v_{k+1}^{(i)} \Delta t, \quad v_{k+1}^{(i)} = \alpha v_k^{(i)} + \beta \frac{x_{\text{best}}^{(i)} - x_k^{(i)}}{\Delta t} + \gamma \frac{x_{\text{best}} - x_k^{(i)}}{\Delta t}.$$

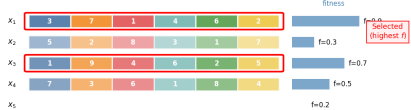
Define all symbols.

Solution:

- $x_k^{(i)}$: position of particle i at iteration k
- $v_k^{(i)}$: velocity of particle i at iteration k
- $x_{\text{best}}^{(i)}$: best position ever visited by particle i (personal best)
- x_{best} : best position ever found by *any* particle in the swarm (global best)
- α : inertia weight (momentum of previous velocity)
- β : cognitive coefficient (attraction to personal best)
- γ : social coefficient (attraction to global best)
- Δt : time step; k = iteration index; i = particle index

1. Selection

Evaluate fitness $f(x_i)$ for each individual. Select parents with probability proportional to fitness (roulette wheel or tournament).



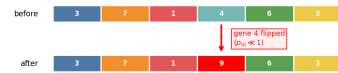
2. Crossover

Pair selected parents. Choose a random cut point and swap tails to produce offspring. (Also: uniform or two-point crossover.)



3. Mutation

With small probability $p_m \ll 1$, randomly change one gene. Prevents premature convergence and maintains genetic diversity.



Adv: global search; no gradient; handles discrete/mixed/discontinuous f

Dis: many f evals; no convergence guarantee; tuning required ($\alpha, \beta, \gamma, p_m$, pop size)

Praxis Quest $p_0 = -\nabla f(x_0)$

$f(x_1, x_2) = x_1^2 + 5x_2^2 \quad x \in \mathbb{R}^2 \quad x_0 = (3, 1)^T$

$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \quad \nabla f(x_0) = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$

$\hat{x}_1 = x_0 + \alpha p_0 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - \alpha \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 3-6\alpha \\ 1-10\alpha \end{bmatrix}$

$\phi(\alpha) = (3-6\alpha)^2 + 5(1-10\alpha)^2$

$= 9 - 36\alpha + 36\alpha^2 + 5(1 - 20\alpha + 100\alpha^2)$

$= 9 - 36\alpha + 36\alpha^2 + 5 - 100\alpha + 500\alpha^2$

$= 14 - 136\alpha + 536\alpha^2$

$d\phi/d\alpha = -136 + 1072\alpha \rightarrow \alpha_0 = \frac{136}{1072} = 0.1269$

$x_1 = x_0 - \alpha_0 \nabla f(x_0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix} - 0.1269 \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} 2.239 \\ 0.7614 \end{bmatrix}$

$s_0 = -\alpha_0 \nabla f(x_0) = -0.1269 \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -0.7614 \\ -1.269 \end{bmatrix}$

$y_0 = \nabla f(x_1) - \nabla f(x_0)$

$y_0 = \begin{bmatrix} 2 \cdot 2.239 \\ 10 \cdot (-0.7614) \end{bmatrix} - \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -1.522 \\ -12.687 \end{bmatrix}$

Gradient Descent

$MUM: f(x_1, x_2) = 2x_1^2 + x_2^2 \quad x_0 = (2, 4)^T$

$\nabla f = \begin{bmatrix} 4x_1 \\ 2x_2 \end{bmatrix} \quad p_0 = -\nabla f = \begin{bmatrix} -8 \\ -8 \end{bmatrix}$

$x_0 + \alpha p_0 = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \alpha \begin{bmatrix} -8 \\ -8 \end{bmatrix} = \begin{bmatrix} 2-8\alpha \\ 4-8\alpha \end{bmatrix}$

$\phi(\alpha) = 2(2-8\alpha)^2 + (4-8\alpha)^2$

$= (4-32\alpha+64\alpha^2) + (16-64\alpha+64\alpha^2)$

$= 8-64\alpha+128\alpha^2$

$\frac{d\phi}{d\alpha} = -64 + 256\alpha \rightarrow \alpha = \frac{64}{256} = 0.25$

$p(p, v) = \frac{1}{2} p^T H(p) p + v^T g(p) + \frac{1}{2} v^T g''(p) v$

$f(x) = x_1^2 + 5x_2^2 \quad g(x) = x_1 + x_2 - 4 \quad x_0 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \quad \nabla g(x) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \nabla g''(x) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$H_f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix}$

$A = \frac{1}{2} p_1^T \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix} p_1 = \frac{1}{2} p_1^T A p_1 = p_1^T A p_1$

$B = \nabla f(x_0) p_1 = \begin{bmatrix} 4 \\ 20 \end{bmatrix} p_1 = 4p_1 + 20p_2$

$C = \nabla g(x) = \nabla(p_1 + p_2) = p_1 + p_2$

$D = \nabla g''(x) (p_1, p_2) = \nabla p_1 + \nabla p_2$

Ansatz

$\Gamma = g(x_0) + \nabla g(x_0) p = p_1 + p_2 - 4 + p_1 + p_2$

Gradient Descent

Iteration 1

$x_1 = x_0 + \alpha p_0 = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} -8 \\ -8 \end{bmatrix} = \begin{bmatrix} 2-4 \\ 4-4 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$

$x_1 = 2 - 2.667\alpha = 0.667$

$\nabla f(x_1) = \begin{bmatrix} 4 \cdot (-2.667) \\ 2 \cdot (-2.667) \end{bmatrix} = \begin{bmatrix} -10.667 \\ -5.333 \end{bmatrix}$

$p_1 = \begin{bmatrix} 10.667 \\ 5.333 \end{bmatrix}$

$x_1 + \alpha p_1 = \begin{bmatrix} 0.667 \\ -2.667 \end{bmatrix} + \alpha \begin{bmatrix} 10.667 \\ 5.333 \end{bmatrix}$

$x_1 = \begin{bmatrix} -0.667 + 10.667\alpha \\ -2.667 + 5.333\alpha \end{bmatrix}$

$\phi(\alpha) = 2(-0.667 + 10.667\alpha)^2 + (1.333 - 2.667\alpha)^2$

$= 2(A^2 + 10AB + 100B^2) + (C^2 + 2CD + D^2)$

$= 2A^2 + C^2 + (4AB + 20C) \alpha + 20B^2 + 2D^2$

$2A^2 + C^2 = 2.667 \quad \phi(\alpha) = 2.667 - 14.667\alpha + 21.333\alpha^2$

$4AB + 20C = 14.667 \quad \frac{d\phi}{d\alpha} = 4.667 - 43.334\alpha$

$20B^2 + D^2 = 21.333 \quad \frac{d\phi}{d\alpha} = 4.667 - 43.334\alpha \rightarrow \alpha = 0.1077$

$f(x) = x_1^2 + 5x_2^2 \quad g(x) = x_1 + x_2 - 4 \quad x_0 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \quad \nabla g(x) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \nabla f(x_0) = \begin{bmatrix} 4 \\ 20 \end{bmatrix}$

$H_f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix} \quad g(x) = 2 + 3.4 = 1$

$A = \frac{1}{2} p_1^T H_f(x) p_1 = \frac{1}{2} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = p_1^T A p_1 + 5p_2^T$

$B = \nabla f(x_0) p = \begin{pmatrix} 4 \\ 20 \end{pmatrix} p = 4p_1 + 20p_2$

$C = \nabla g(x_0) = \nabla = p_1 + p_2$

$D = \nabla g''(x_0) p = \nabla \begin{pmatrix} 1 \\ 1 \end{pmatrix} p = \nabla p_1 + \nabla p_2$

$\Gamma = g(x_0) + \nabla g(x_0) p = 1 + \begin{pmatrix} 1 \\ 1 \end{pmatrix} p = 1 + p_1 + p_2$

$\frac{d}{d\alpha} \Gamma = p_1^T A p_1 + 4p_1 + 20p_2 + p_1 + p_2$

$\Gamma = 1 + p_1 + p_2$